2. Observe the above completed Pascal's Triangle and moving the slanting strips, find the sequence that you see in it and complete them. One is done for you



(i) 1, 2, 3, 4, 5, 6, 7.

(ii) 1, 3, ___, ___, ___, ___.

(iii) 1, ___, ___, ___, ___.

(iv) ___, ___, ___, ___.

3. Observe the sequence of numbers obtained in the 3rd and 4th slanting rows of Pascal's Triangle and find the difference between the consecutive numbers and complete the table given below.

(i)

3rd slanting row	1	3	6	10	15	21
Difference	2	___	4	___	6	

(ii)

4th slanting row	1	4	10	20	35
Difference	3	___	10	___	

Example 5.2

Tabulate the 3rd slanting row of the Pascal's Triangle by taking the position of the numbers in the slanting row as x and the corresponding values as y.

x	1	2	3	4	5	6	...
y	1	3	6	10	15	21	...

Verify whether the relationship, $y = \frac{x(x+1)}{2}$ between x and y for the given values is true.

Solution

Observe the table carefully. To verify the relationship between x and y, let us substitute the values of x and get the values of y.

If $x = 1$, $y = \frac{1(1+1)}{2} = \frac{2}{2} = 1$

If $x = 2$, $y = \frac{2(2+1)}{2} = \frac{6}{2} = 3$

If $x = 3$, $y = \frac{3(3+1)}{2} = \frac{12}{2} = 6$

Think

The values of y are obtained by half of the product of the two consecutive values of x.

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